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MEDICATION CARTRIDGE

Abstract

The present disclosure provides a medication cartridge and associated dispensing device. The dispensing device includes a housing containing a cartridge carrier, designed with a specific connector to align with a medication cartridge. The medication cartridge includes a shell with a medicine vial slidably positioned therein. When the dispensing device is activated, the cartridge carrier rotates to raise the vial and cause the vial to dispense medicine from the vial and out of a dispensing aperture for use by a patient.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a Continuation-in-Part application of U.S. Serial No. U.S. Ser. No. 18/600,198, filed Mar. 8, 2024, the entirety of which is incorporated herein by reference. [0002] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/600,402, filed Mar. 8, 2024, the entirety of which is incorporated herein by reference. [0003] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/601,329, filed Mar. 11, 2024, the entirety of which is incorporated herein by reference. [0004] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/655,722, filed May 6, 2024, which is a Continuation application of U.S. Ser. No. 18/606,136, filed Mar. 15, 2024, now issued as U.S. Pat. No. 12,106,836, the entirety of which are incorporated herein by reference. [0005] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/606,151, filed Mar. 15, 2024, the entirety of which is incorporated herein by reference. [0006] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/606,173, filed Mar. 15, 2024, the entirety of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) Field of Invention

[0007] The present invention relates to a medication dispensing system and, more particularly, to a mechanized medication cartridge and associated handheld oral medication dispensing device for actuating the medication cartridge.

(2) Description of Related Art

[0008] The field of medication treatment has grown increasingly complex in recent years, with patients frequently needing to manage and administer multiple types of medication. Under such circumstances, it is vital to provide reliable, accessible, and simple methodologies to patients for dispensing their medications. Many existing dispensing devices, however, tend to be cumbersome, confusing, or inefficient. Often, they rely on patients having to manually change their medication reminders or products, which can involve complicated steps or create potential for mistakes, like grabbing the wrong medication or dispensing an incorrect dosage. Furthermore, mistakes may potentially have serious health implications. Hence, it's vital to design an easy-to-use and fail-safe medication dispensing device.

[0009] Further, existing oral medications are typically taken as tablets or syrups. Such medication delivery items may be functional, but often take time for efficacy due to digestive and other body related functions. Additionally, such medication delivery techniques are inconvenient to carry on one's person or inconspicuously use. Thus, a need exists for a unique oral medication dispensing

device that provides an alternative to tablet and syrup-type medications.

[0010] One embodiment disclosed herein pertains to a handheld oral medication dispensing device. To provide for ease of use and reliability, the dispensing device is formed with a mechanized pump. The purpose of this design is not only to make medication administration simpler and more efficient but also to mitigate potential errors between users. The innovative design of a mechanized pump in the dispensing device in conjunction with a keyed connector and the configuration of the cartridges can significantly aid patients to manage and administer their medication effectively and safely.

SUMMARY OF INVENTION

[0011] The present disclosure is directed to a medication cartridge that is operable with an associated dispensing device. The medication cartridge comprises a shell having a dispensing aperture and a vial slidably positioned within the shell, the vial having a pump mechanism with an outlet aligned with the dispensing aperture. The vial is operable for rising within the shell to activate the pump mechanism and dispense medicine from the vial and dispensing aperture.

[0012] In another aspect, the shell includes one or more rising slots passing through the shell and rising from a bottom end of the shell toward an apex.

[0013] In another aspect, the vial includes one or more rising channels that align with the one or more rising slots and rise from a bottom end of the vial toward the apex.

[0014] In another aspect, the shell includes one or more lateral slots that traverse laterally away from the apex.

[0015] In yet another aspect, the vial includes downward descending channels that descend downward at an angle from the apex.

[0016] In yet another aspect, the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.

[0017] Further, the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.

[0018] In yet another aspect, the vial is slidable within the shell between a stored position and a full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.

[0019] Additionally, the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.

[0020] Finally, as can be appreciated by one in the art, the present invention also comprises a method for forming and using the invention as described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The objects, features and advantages of the present invention will be apparent from the following detailed descriptions of the various aspects of the invention in conjunction with reference to the following drawings, where:

[0022] FIG. 1A is an illustration of a first medication cartridge according to a first embodiment;

[0023] FIG. 1B is an illustration of a second medication cartridge according to a second embodiment;

[0024] FIG. 1C shows an exemplary illustration a handheld medication dispensing device and the first medication cartridge according to the first embodiment;

[0025] FIG. 2A shows an interior-view of the medication dispensing device, depicting a cartridge

carrier;

[0026] FIG. 2B shows an interior-view of the medication dispensing device, depicting the first medication cartridge inserted into the cartridge carrier;

[0027] FIG. 3 is an interior-view illustration of the medication dispensing device, depicting an activator mechanism as operably connected with the cartridge carrier;

[0028] FIG. 4 shows an exemplary illustration of the cartridge carrier and first medication cartridge;

[0029] FIG. 5A shows a side-view illustration of the first medication cartridge;

[0030] FIG. 5B shows a top-view illustration of the first medication cartridge;

[0031] FIG. 5C is a cross-sectional, side-view illustration of the first medication cartridge, taken from line A-A of FIG. 5B;

[0032] FIG. 6 shows an exploded-view illustration of the first medication cartridge;

[0033] FIG. 7A shows a front-view illustration of the first medication cartridge;

[0034] FIG. 7B shows a right, side-view illustration of the first medication cartridge;

[0035] FIG. 7C shows a left, side-view illustration of the first medication cartridge;

[0036] FIG. 7D shows a rear-view illustration of the first medication cartridge;

[0037] FIG. 7E shows a top-view illustration of the first medication cartridge;

[0038] FIG. 7F is a cross-sectional, side-view illustration of the first medication cartridge, taken from line A-A of FIG. 7C;

[0039] FIG. 7G is an illustration of a lotion pump assembly of the prior art;

[0040] FIG. 7H is an illustration of an example pump assembly as applied to the first medication cartridge in accordance with various embodiments of the present invention;

[0041] FIG. 8A shows an exemplary illustration of the cartridge carrier and the first medication cartridge;

[0042] FIG. 8B shows an exemplary illustration of the cartridge carrier and the first medication cartridge, depicting the first medication cartridge as being inserted into the cartridge carrier;

[0043] FIG. 9 shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier;

[0044] FIG. 10A shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the first medication cartridge in a stored position;

[0045] FIG. 10B shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier rotating to move the first medication cartridge up and toward the dispensing position;

[0046] FIG. 10C shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier as rotated to move the first medication cartridge up and into the dispensing position;

[0047] FIG. 11A shows an exemplary illustration of the first medication cartridge, depicting the vial sliding within the shell;

[0048] FIG. 11B shows close-up view of Detail A, taken from FIG. 11A;

[0049] FIG. 11C shows an exemplary illustration of the first medication cartridge, depicting the vial within the shell;

[0050] FIG. 12A shows an exemplary illustration of the first medication cartridge within the cartridge carrier, depicting the first medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the first medication cartridge;

[0051] FIG. 12B shows an exemplary illustration of the first medication cartridge within the cartridge carrier, depicting the first medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the first medication cartridge;

[0052] FIG. 13A is an interior-view illustration of the medication dispensing device;

[0053] FIG. 13B is an interior-view illustration of the cartridge carrier;
[0054] FIG. 14A is an exterior-view illustration of the second medication cartridge;
[0055] FIG. 14B is an interior-view illustration of the second medication cartridge;
[0056] FIG. 14C is an interior-view illustration of the second medication cartridge, depicting a cartridge shell removed for illustrative purposes to depict the medication vial therein;
[0057] FIG. 15A is an illustration on the second medication cartridge and the handheld medication dispensing device;
[0058] FIG. 15B is an illustration depicting the second medication cartridge being inserted into the handheld medication dispensing device;
[0059] FIG. 16A is an illustration depicting the second medication cartridge in a stored position;
[0060] FIG. 16B is an illustration depicting the second medication cartridge being actuated in an intermediate position to raise the vial and pump medicine from the vial;
[0061] FIG. 16C is an illustration depicting the second medication cartridge being actuated in a full pump position, with the vial completely raised to pump medicine from the vial;
[0062] FIG. 16D is an illustration depicting the second medication cartridge in an intermediate position as it returns to a stored position and lowers the vial; and
[0063] FIG. 16E is an illustration depicting the second medication cartridge in the stored position.

DETAILED DESCRIPTION

[0064] The present invention relates to a medication dispensing system and, more particularly, to a mechanized medication cartridge and associated handheld oral medication dispensing device for actuating the medication cartridge. The following description is presented to enable one of ordinary skill in the art to make and use the invention and to incorporate it in the context of particular applications. Various modifications, as well as a variety of uses in different applications will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to a wide range of embodiments. Thus, the present invention is not intended to be limited to the embodiments presented, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0065] In the following detailed description, numerous specific details are set forth in order to provide a more thorough understanding of the present invention. However, it will be apparent to one skilled in the art that the present invention may be practiced without necessarily being limited to these specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the present invention.

[0066] The reader's attention is directed to all papers and documents which are filed concurrently with this specification and which are open to public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference. All the features disclosed in this specification, (including any accompanying claims, abstract, and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is only one example of a generic series of equivalent or similar features.

[0067] Furthermore, any element in a claim that does not explicitly state "means for" performing a specified function, or "step for" performing a specific function, is not to be interpreted as a "means" or "step" clause as specified in 35 U.S.C. Section 112, Paragraph 6. In particular, the use of "step of" or "act of" in the claims herein is not intended to invoke the provisions of 35 U.S.C. 112, Paragraph 6.

[0068] Please note, if used, the labels left, right, front, back, top, bottom, forward, reverse, clockwise and counter clockwise have been used for convenience purposes only and are not intended to imply any particular fixed direction. Instead, they are used to reflect relative locations and/or directions between various portions of an object.

(1) Description

[0069] As noted above, the present invention relates to a medication dispensing system having a

mechanized medication cartridge and, in some aspects, an associated handheld oral medication dispensing device for actuating the medication cartridge. This disclosure provides two example embodiments of the medication cartridge, both of which can be used with the handheld medication dispensing device. FIG. 1A depicts a first embodiment of the medication cartridge **102** while FIG. 1B depicts a second embodiment of the medication cartridge. It should be noted that handheld medication dispensing device as described herein can be programed to work with either medication cartridge, as well as other designs as can be appreciated by those skilled in the art. For clarity, FIGS. 1C through 12B are described and illustrated with respect to the first embodiment of the medication cartridge, while FIGS. 13A through 16E are described and illustrated with respect to the second embodiment of the medication cartridge.

[0070] As shown in FIG. 1C, the handheld medication dispensing device **100** is designed primarily to securely store and dispense medication from a medication cartridge **102**. The device **100** comprises a housing **104**, principally structured to protect the critical internal components while offering ergonomic suitability for handheld operation. The housing **104** exterior is designed with both convenience and comfort in mind, allowing a user to hold the device **100** and access the medication with ease. In an exemplary embodiment, the housing **104** may be configured in a variety of shapes, sizes, and materials to accommodate different user preferences and medication types.

[0071] In some aspects the device **100** includes an authentication component **106** that is designed to authenticate a user's identification in various manners and, upon authentication, allow for operation and control of the device **100**. The authentication component **106** is any suitable mechanism, device, system, etc., that allows for user authentication, non-limiting examples of which include a fingerprint reader built into the device **100** or facial recognition via a camera and app, etc. that communicates with the device upon authentication to notify the device **100** of authentication and allow for operation of the device **100**, or any other known authentication method or device. The authentication component **106** is set up to permit the authenticated or designated user access to the operation of the said device **100**, thereby ensuring a highly secure and personal way of dispensing medication. In one embodiment and as illustrated, the authentication component **106** includes a fingerprint reader and all associated components as necessary to allow for authentication with such a reader, including an integrated circuit, power source (battery), etc. For example, in a first use, the authentication component **106** can be configured to recognize the user and, thereafter, only allow access to the recognized user to cause the device **100** to operate as intended. Such a setup can be configured in any suitable manner as understood by those skilled in the art. As non-limiting example, a first use may require that a user turn on the device **100**, which provides indicia (e.g., via a light, digital screen, etc.) to notify the user to place their finger on the fingerprint reader. The fingerprint reader then reads and stores the fingerprint of the designated user. Thereafter, the activator mechanism can only be activated by the designated user upon placement of the finger on the fingerprint reader. In another example, software or a phone application can be used to wirelessly communicate (e.g., via Wi-Fi, Bluetooth, etc.) with the device **100** to program the device **100** to only recognize the designated user via the authentication component **106**. Thus, in some aspects, the device **100** includes all necessary components to allow it to wirelessly communicate with an external device, non-limiting examples of which include Wi-Fi transceivers/components, Bluetooth components/gear, etc. In an alternative embodiment, the device **100** can simply be turned off and on and includes a button or other similar feature to cause the device **100** to operate and actuate the activator mechanism after user authentication, as described in further detail below.

[0072] In another embodiment and in the case of facial recognition via a camera and app, etc. on a mobile phone that communicates with the device **100**, the authentication component **106** includes components housed within device **100** to allow for remote authentication, including the programming on an integrated circuit or similar device, along with the wireless communication

components necessary to communication with the mobile phone (e.g., Bluetooth, WiFi transceivers, etc.) to allow for authentication and operation of the device **100**. The software or application that is downloaded onto the mobile phone or device includes the programming and any other necessary information to allow for user authentication and cause the authentication to be transmitted and received by the authentication component **106**.

[0073] In one aspect, the authentication component **106** is a biometric fingerprint reader linked to a mobile app, which can be used to prevent unauthorized medication dispensing, requiring user authorization via the app for initialization, and connecting to caregivers and personal networks to ensure medication adherence. The design of the device **100**, accessories, and app is centered around the need for an easier and better way to receive the right dose, at the right time, in the right way and frequency. In one aspect, controlled via the app, the handheld device **100** can alert users to their medication schedule, while also notifying caregivers and family members about adherence. For restricted medications, the device **100** and associated app can restrict dosing until authorized by the prescribing physician. Tampering with the medication cartridge **102** triggers alerts to caregivers, pharmacists, and physicians, ensuring safe usage, especially for restricted medications by alerting through the app that the medication has not been taken as prescribed. In another aspect, the device **100** and medication cartridge **102** are formed to include any necessary components that allow the device **100** to recognize a specific medication cartridge **102**. As a non-limiting example, each medication cartridge **102** can be formed to include a Radio Frequency Identification (RFID) tag, while the device **100** includes an RFID reader (or other similar technology).

[0074] As noted above, in some aspects, the device **100** is formed to wirelessly communicate with an external device (e.g., mobile phone) and its app to allow for the external device to control any features as applicable to the device **100**. In doing so, the dispensing device **100** includes a wireless communication component (depicted as element **202** in FIG. 2) which include all necessary components to allow it to wirelessly communicate with the external device and control the dispensing device **100** to provide the desired operations, non-limiting examples of which include Wi-Fi transceivers/components, Bluetooth components/gear (e.g., a system on a chip (SoC) that includes a transceiver, antenna and control chip), integrated circuit(s), a power source (e.g., battery), and/or any another other electronic components as may be needed to allow for wireless or Bluetooth connectivity and related operations of the device **100**. In one aspect and per Bluetooth protocol, when the dispensing device **100** links with the associated mobile external device to communicate, they form an ad hoc mini computer network referred to as a piconet. Within the piconet, the external device, referred to as the master, assumes the control of the network, issuing instructions to the dispensing device **100**, which is referred to as a slave to control a variety of features and operations.

[0075] As noted above, several programmable features include controlling user access and operation of the device **100** (and activator mechanism, etc.) based on user authentication, dosing schedules, restricting dosing until authorized by a prescribing physician, etc. As a non-limiting example, after a medication cartridge **102** is inserted into the device **100**, it is essentially locked and not allowed to position the medication cartridge **102** into the dispensing position until access is granted. This access can be based on the mobile phone user authentication process which transmits an “access granted” signal (via Bluetooth, etc.) to the device **100** and allows the activator mechanism (described in further detail below) to operate and move the medication cartridge **102** to the dispensing position. Without such an “access granted” signal that is wirelessly received by the dispensing device **100**, the device **100** is locked and will not activate the activator mechanism. As yet another non-limiting example, the dosing schedule as recorded into the software application and external device may specify that the medication is be dispensed a certain number of times daily. A user would then connect their mobile external device (e.g., mobile phone) via Bluetooth to the dispensing device **100** during the allotted times, at which point an “access granted” signal is provided to the dispensing device **100**. The software application as stored in the mobile external

device will only provide the “access granted” signal that corresponds to the dosing schedule, while the dispensing device **100** is locked or deactivated during all other times. In other words and as can be appreciated by those skilled in the art, there are a variety of features of the dispensing device **100** that can be operated by providing the wireless communication components that allow for wireless control of the dispensing device **100** via a mobile phone or other external wireless device. [0076] As shown in the interior views of FIGS. 2A and 2B, integrated within the housing **104** is a cartridge carrier **200**. The cartridge carrier **200** functions as a receptacle for the medication cartridge **102**. The cartridge carrier **200** includes a keyed connector, most importantly, innovatively designed to align with a specified type of medication cartridge **102**. This keyed connector serves as an interlocking structure that carefully guides the positioning and orientation of the cartridge **102**, allowing only for the insertion of a compatible medication cartridge **102**. The keyed connector along with the cartridge carrier **200** ensure the correct placement and alignment of the medication cartridge **102**, which can help eliminate potential misuse or cartridge damage, and fostering a smooth transition between stored and dispensing positions (as described in further detail below). [0077] As shown in the interior view of FIG. 3, an activator mechanism **300** forms part of the configuration and is integrated within the housing **104**. The activator mechanism **300**, when engaged or otherwise activated, prompts the cartridge carrier **200** to actuate. The involvement of the activator mechanism **300** ensures a controlled movement of the inserted medication cartridge **102** between at least two positions—a stored position and a dispensing position. Note that FIG. 2A depicts the medication cartridge in a stored position, while FIG. 3 depicts the medication cartridge in a dispensing position.

[0078] In the stored position, as shown in FIG. 2A, the medication is kept secure within the device **100**. The position ensures safety by preventing unintentional discharge of medication when not in use. When the activator mechanism **300** is activated, the cartridge carrier **200** rotates to lift up the medication cartridge **102** into the dispensing position. When this dispensing position is selected, the medication cartridge **102** is correctly oriented to dispense the medication through a dispensing aperture (described in further detail below) integrated into the medication cartridge **102**.

[0079] The link between the activator mechanism **300** and the cartridge carrier **200** allows for reliable and essential movement control. This means the user can confidently operate the device **100** knowing the medication is dispensed accurately and safely.

[0080] The cartridge carrier **200** is formed in any suitable manner to allow for activation by the activator mechanism **300**, which results in transitioning the medication cartridge **102** between the stored and dispensing positions. In an embodiment and as shown in FIG. 3, the cartridge carrier **200** is rotatable within the housing **104**. The cartridge carrier **200** may be formed of materials that possess strength and durability such as, but not limited to, metals, polymers, and/or composites. The size, shape, and dimensions of the cartridge carrier may vary as per the application requirements.

[0081] The housing **104** may be constructed to accommodate the rotatable cartridge carrier **200**. The construction of the housing **104** can also encompass different types of materials that provide the necessary strength and durability inclusive of, but not limited to metals, polymers, composites, among others. The interior of the housing **104** is ideally designed to allow the cartridge carrier **200** to rotate through actuation of the activator mechanism **300**.

[0082] The activator mechanism **300**, ensuring the cartridge carrier's **200** rotation within the housing **104**, can be implemented through a variety of means, such as mechanical gears, electric motors, a battery, manually operated mechanisms, integrated circuits, or any combination thereof. This activator mechanism **300** administration permits the user to precisely position the cartridge carrier **200** within the housing **104** between a first position (in which the medication cartridge **102** is inserted into the cartridge carrier **200** and is stored in a stored position) and a second position (in which the medication cartridge **102** is lifted from the housing **104** to a dispensing position). The activator mechanism **300** can then be activated to rotate the cartridge carrier **200** in an opposite

direction to return the medication cartridge **102** to the stored position.

[0083] In one aspect, the dispensing device **100** includes a locking mechanism that is configured to selectively lock/unlock the medication cartridge **102** within the cartridge carrier **200**. The locking mechanism may involve an interlocking system, a magnetic attachment, or any other secure attachment mechanism. As a non-limiting example, the locking mechanism may be a spring-loaded detent style 'click-in' and 'click-out' mechanism formed at the bottom of the medication cartridge **102** and cartridge carrier **200**. In another aspect, the locking mechanism can be an electronically actuated lock (e.g., mechanized tab, etc.) that locks the medication cartridge **102** into the cartridge carrier **200**. For example, the exterior surface of the medication cartridge can include a slot that is formed to accommodate a tab that is electronically moved (e.g., turned/slid, etc.) into the slot when locking and, in the alternative, pulled from the slot when released. As yet another non-limiting example, the locking mechanism can include programming such that after the cartridge **102** is inserted into the cartridge carrier **202**, the cartridge carrier **200** is rotate partially until alignment features (described in further detail below and depicted as element **406**) are positioned in the middle of the second slot (described in further detail below and depicted as element **710**), with the cartridge carrier **200** then stopped until it is desired to rotate further to position the medication cartridge **102** in the dispensing position. With the alignment features in the middle of the second slot, the cartridge **102** is effectively locked within the cartridge carrier **200**. Thus, as can be appreciated by those skilled in the art, there are several mechanisms that can be employed to selectively lock the medication cartridge **102** into the cartridge carrier **200**.

[0084] In another aspect, the inclusion of the locking mechanism may also be included to secure the cartridge carrier **200** at any desired rotational position, thus preventing unintentional and unwanted movement when the cartridge carrier **200** is set in a particular position. This locking mechanism can be integrated into the cartridge carrier **200**, the housing **104**, the activator mechanism **300**, or combinations of these components. This rotatable cartridge carrier **200** system installed within a housing **104** assembly may have wide ranging applications, such as, for instance, accommodating various cartridges for providing users with quick and easy access to different medications and dosages, multiple configurations, interchangeability of components, and more. Thus, the present one embodiment discloses a rotatable cartridge carrier **200** system, promoting ease of use, enhancing performance, providing efficient operation, flexibility, improved control and precision, leading to a more efficient and user-friendly functionality.

[0085] In one embodiment and as shown in FIG. 4, the cartridge carrier **200** includes both a proximal end **400** and a distal end **402**. The proximal end **400** of this unit is open, and it is specifically structured to receive the medication cartridge **102**. This can be any kind of medication cartridge **102**, and the open design simplifies the process of interchanging cartridges of differing medications, volumes, or other specifications, thus making it broadly applicable and versatile.

[0086] The distal end **402** of the cartridge carrier **200** includes a key component of this one embodiment—a gear wheel **404**. This gear wheel **404** is not a mere inclusion but is operably connected to the activator mechanism **300**. The gear wheel **404** and activator mechanism **300** are conceptually designed to work in harmony. In operation, when the user deploys the activator mechanism **300**, the included gear wheel **404** is rotated, thereby rotating the cartridge carrier **200**.

[0087] The gear wheel **404** is a significant element in the functioning of this device **100** mainly due to its interaction with the activator mechanism **300**. The gear wheel **404**, upon the interaction with activator mechanism **300**, contributes to the precision with which the cartridge carrier **200** is rotated to position the medication cartridge **102** between the stored and dispensing positions. Its size, teeth count, and interaction with other gears if connected, contribute to its function.

[0088] Moreover, the position of the gear wheel **404** at the distal end **402** of the cartridge carrier **200** may also assist in a user-friendly, ergonomic design. The gear wheel's **404** location ensures a seamless connection between the mechanical parts of the activator mechanism **300** and the carrier cartridge **200**, augmenting the device's **100** overall performance.

[0089] As noted above and referring again to FIG. 3, the device **100** includes an activator mechanism **300** to selectively turn the gear wheel **404** and, by extension, rotate the cartridge carrier **200**. The activator mechanism **300** includes any components as necessary to rotate the gear wheel **404**. In one embodiment, the activator mechanism **300** comprises of at least a power source (e.g., battery **304**), a motor **302**, and one or more gears **306** that are operably connected to the gear wheel **404**. The battery **304** in the device **100** forms the power source, capable of providing a stable and reliable supply of power to the other components. The size, capacity, and type of the battery **304** may be customized according to design preferences, device requirements, or various other factors, while being compliant with standard safety and operation guidelines. In some aspects, the battery **304** is encased within the device **100** and intended for device **100** replacement upon depletion of the battery **304**, while in other aspects, it is replaceable. In yet other aspects, the battery **304** is rechargeable using a charging port **308** or any other charging means as known to those skilled in the art.

[0090] The motor **302** functions as the primary mechanical component that is responsible for movement or force in the device **100**. Upon activation, the motor **302** draws power from the battery and begins to turn to rotate any of the one or more gears and the operably connected gear wheel **404**. The type, size, and characteristics of the motor **302** can be adjusted based on the requirements of the device, anticipated load, desired performance factors among other parameters.

[0091] The one or more gears represent a transmission mechanism, operably connected to the motor **302** and the gear wheel **404**. Upon the activation of the device **100**, the rotational force produced by the motor **302** is transferred to these gears. The gears enable the transfer of power from the motor **302** to the gear wheel **404**. Upon receiving the force from the one or more gears of the activator mechanism **300**, the gear wheel **404** begins to turn. This turning of the gear wheel **404** then contributes to the rotation of a cartridge carrier **200**.

[0092] Referring again to FIG. 4, one embodiment pertains to an enhanced cartridge carrier **200** system, specifically designed with one or more alignment features **406** that protrude into the cartridge carrier **200** for a keyed connection and accurate cartridge **102** positioning, thereby improving the overall functionality of the cartridge **102** and the cartridge carrier **200** system.

[0093] The alignment features **406** of one embodiment are distinguishable protrusions present on the inner surface of the cartridge carrier **200**. These alignment features **406** serve to guide the cartridge **102** into the correct position within the cartridge carrier **200** when installing a cartridge **102** into the system. These protruding alignment features **406** also help to prevent misalignment and displacement of the cartridge **102** while it is installed and during operation. Notably and as described below in further detail, the alignment features **406** are critical in transitioning an inserted cartridge **102** between the stored and dispensing position.

[0094] The alignment features **406** are strategically located within the cartridge carrier **200** in a manner to engage with corresponding features or portions on the cartridge **102**. When the cartridge **102** is inserted into the carrier **200**, the protruding alignment features **406** guide it into the correct position as the features **406** fit into compatible sections (i.e., alignment channels **408**) of the cartridge **102**. This results in a snug and correct fit of the cartridge **102** within the cartridge carrier **200**.

[0095] Furthermore, these alignment features **406** may come in various shapes and sizes and can be constructed using various materials that are suitable for the system and the intended application. They can be rigid to withstand the pressure exerted by the cartridge **102** as it is inserted into the cartridge carrier **200**, or flexible to allow for slight variations in cartridge **102** size and shape.

[0096] For further understanding, FIGS. 5A through 5C depict side, top, and cross-sectional views, respectively, of the cartridge carrier **200** and alignment features **406**. These features **406**, protruding into the interior of the cartridge carrier **200**, support the accurate placement of cartridges **102** within the device. Thus, the functionality and efficiency of the overall dispensing device **100** are significantly improved along with consistent cartridge **102** performance ensured by these alignment

features **406**. This inventive cartridge carrier **200** system design genuinely adds value to applications where precise alignment of cartridges **102** in their carrier systems is crucial. This one embodiment indeed ensures an innovative update for improving the efficiency of cartridge insertion and alignment in cartridge carrier systems, potentially paving the way for advancements in devices utilizing such systems.

[0097] As noted above and referring again to FIG. **4**, the present disclosure also provides a unique medication cartridge **102** that can be inserted within a cartridge carrier, thus providing a novel means to carry, store and administer medication. The essence of this one embodiment lies in the exclusive design of the medication cartridge **102** and the cartridge carrier that is not only easy to operate but also provides secure storage and effortless administration of the medication.

[0098] The medication cartridge **102** is an integral part of the inventive concept and is designed to contain distinct types of medications, be it in liquid, gel, or cream, or other dispensable forms of medication. The cartridge **102** can be made up of pharmaceutically safe materials ensuring no chemical reactions occur when in contact with the medication. The cartridge **102**, itself can showcase cylindrical or other geometrical shapes as required by shape of the cartridge carrier **200**. The medication cartridge **102** and cartridge carrier **200** partnership provides an ideal solution in various medical environments, such as hospitals, pharmacies, nursing homes, or even domestic settings. It proves to be beneficial where multiple medications should be stored and administered safely and hygienically. It combines innovation with convenience, improving medication storage, and administration methodologies.

[0099] As noted above and as shown in FIG. **6**, the cartridge **102** itself is also designed to provide a marked improvement over other medication containers. It is envisioned to include markings or identification systems for easily recognizing the type of medication stored within each cartridge **102**. The real innovation lies in the incorporation of one or more channels **408** and aligned slots **606** within the cartridge **102**. These are constructed meticulously within the cartridge **102** and serve a distinct function of aligning and receiving one or more alignment features **406**. As will be evident below, these channels **408** and slots **606** engage with the aforementioned alignment features **406** to allow for controlled operation of the device **100**.

[0100] Further and as shown throughout FIGS. **6** through **7F**, the present disclosure provides an improved configuration for a medication cartridge **102**, particularly advantageous for medications that require enhanced storage for securely maintaining their efficacy and quality. The innovative design of the medication cartridge **102** includes a robust shell **600** that has a dispensing aperture **604** designed for the release of the medication stored within a medication vial **602**.

[0101] The shell **600**, formed from long-lasting and reliable material (e.g., plastic, metal, etc.), is shaped to house the vial **602** securely and efficiently. This shell **600** acts as a protective layer for the vial **602** placed inside, while maintaining the required conditions for the preservation of the medication. The dispensing aperture **604** deployed on the shell **600** is engineered in a way to align with the outlet **702** of the vial **602**, ensuring precise and mess-free dispensation of the medication.

[0102] Inside the shell **600** is a medication vial **602**, which is shaped and positioned in a manner that it can slide without difficulty within the shell **600**. This ability of the vial **602** to slide enables convenient load and unload of the medication. The vial **602** includes a pump assembly **700** that is equipped with an outlet **702**. The design and positioning of this outlet **702** are such that it aligns perfectly with the dispensing aperture **604** on the shell.

[0103] The pump assembly **700** is any suitable assembly that can be affixed with a container (i.e., the vial **602**) to allow for pumping or otherwise dispensing a medication or fluid therefrom, a non-limiting example of which includes a compression or spring-loaded pump, similar to the pump mechanism as used in a standard lotion bottle. For reference, FIG. **7G** provides an illustration of an example pump assembly as provided for in the prior art, while FIG. **7H** provides a non-limiting example of the pump assembly as modified and applied to the present invention. The pump assembly of the prior art (as shown in FIG. **7G**) includes a clamp **1**, press head **2**, press lever **3**, coat

4, screw cap 5, connect cap 6, gasket 7, piston 8, piston seat 9, spring 10, spring seat 11, valve 12, housing 13, and a straw 14. As applied to the present invention (as depicted in FIGS. 7F and 7H), the pump assembly 700 is connected to the vial 602, while the shell 602 serves as the press head that is used to press down the piston 8 to force the fluid through the straw 14 and out of the dispensing aperture 604. In some instances the pump assembly 700 can be screwed onto the vial 602; however, it should be understood that the invention is not intended to be limited thereto as the pump assembly 700 can be attached in any suitable manner between the vial 602 and shell 600 to allow for dispensing fluid from the vial 602 and out of the dispensing aperture 604. Further, it should be understood that the specific pump assembly 700 as depicted in FIG. 7H is provided as a non-limiting example of a suitable pump assembly and that the invention is not intended to be limited thereto as other mechanisms or assemblies can be used to dispense the fluid upon actuation. For example, the pump assembly can be formed as an airless pump, a peristaltic pump, a diaphragm pump, a pressurized canister style pump, or any other configuration in which activation of the assembly causes the fluid to dispense.

[0104] For illustrative purposes in comparison to a lotion bottle, the vial 602 would serve as the container to hold the fluid, while the shell 600 serves as the pump head that can be depressed to pump the fluid, with the various components of the pump assembly 700 positioned therebetween. In operation and when the medication cartridge 102 is moved to the dispensing position, the alignment features 406 operate to hold the vial 602 in place (due to their locking position in the channel terminal as described in further detail below), while a user can selective press the shell 600 downward around the vial 602 to activate the pump assembly 700 and force fluid from the vial 602.

[0105] For example, the vial 602 includes a reservoir 720 in which the medication is stored. The medication can be in a gas or fluid form; however, desirably and in one aspect, the medication is fluid and can be squirted from the medication cartridge 102 (and vial 602) via actuation of the pump assembly 700. As apparent from the description further below, the pump assembly 700 can be easily operated to cause the medication to be dispensed from the vial 602 through its outlet 702. When a force is applied on the pump assembly 700, it triggers the release of the medication inside of the vial 602, causing the medication to travel from the vial 602, passing through the aligned outlet 702 and dispensing aperture 604, where it is released to the user.

[0106] As noted above, the pump assembly 700 is any suitable assembly, mechanism, or device that can be affixed with the vial 602 to allow for selective dispensing therefrom. As can be appreciated by those skilled in the art, a non-limiting example of such a pump assembly 700 is a spring-loaded system utilizing a stacked one-way valve system to temporarily create a vacuum, allowing fluid to be aspirated from the reservoir 720 and dispensed through the outlet 702 and dispensing aperture 604 upon depression of the shell 600 around the vial 602.

[0107] As noted above, the medication cartridge 102 is generally formed of two components, a shell 600 and a medication vial 602. The vial 602 is sized to be secured within and slide within the shell 600. As shown, the medication cartridge 102 includes a top end 704 and a bottom end 706. Although the shell 600 in FIG. 6 is illustrated as having a cap component 608 at the top end 704, the invention is not intended to be limited thereto as the shell 600 can be formed such that the cap component 608 is integrally formed with the shell 600 as a single unit. In such an aspect, a hole (e.g., sealable hole) can be formed at the bottom end 706 (or top end 704) of the shell 600 to allow for insertion of the vial 602 into the shell 600.

[0108] Notably and of particular importance, the vial 602 includes channels 408 that are aligned with the slots 606 on the shell 600. These channels 408 and slots 606 are built in a manner to accurately align with the alignment features 406 of the cartridge carrier 200. The number and size of these channels 408 and slots 606 can vary depending on the number of alignment features 406 intended to be placed therein. These channels are purposefully built to receive these alignment features, allowing secure and precise positioning. This design ensures that the alignment features do not shift their placement, further guaranteeing the stabilization of the medication within the

cartridge.

[0109] For example, one example embodiment includes a structure involving a pair of opposing channels **408** are formed on opposite sides of the vial **602**. These channels **408** are typically parallel to each other, though variations can occur. In one aspect, each of the pair of opposing channels **408** originates from the lowermost end or bottom end **706** of the vial **602**. The pair of opposing channels **408** rise upward from this point, extending towards the top end **704** of the vial **602**. This ascension continues until they reach a point known as the first junction **714**. It must be clarified that this first junction **714** is not an arbitrary point but is a strategically designed and located structural feature within the device to precisely position the cartridge **102** between stored and dispensing positions.

[0110] Once these channels **408** have reached the first junction **714**, the trajectory shifts. At this point, the channels **408** do not continue with the upward gradient. Instead, they commence a downward traversal toward the bottom end **706**. They proceed downwards, but not vertically downwards. This descent is carried out on an angle around the vial **602**, intended to enhance function and performance.

[0111] These angled downward channels **408** continue to move in the said direction until they reach an area deemed as the second junction **716**. It is at this point that these channels **408** cease to continue. They terminate at this juncture, each channel **408** ending in what is referred to as a channel terminal **718**. As will be apparent below, this distinctive configuration of the channels **708** enables them to perform their intended purpose more efficiently. It is important to consider the precise and specific architectural design of these channels **408**, including their starting point at the bottom end **706** of the vial **602**, their rise towards the first junction **714**, their angled descent towards the second junction **716**, which with respect to the channels, is the final termination at the channel terminal **718**. These particular design elements reveal a thoughtful consideration of their role within the device and contribute both independently and collectively to the overall functionality and performance of the device or system in question.

[0112] The present disclosure also provides an embodiment which is characterized by a pair of opposing slots **606** in the shell **600** that align with the pair of opposing channels **408**. More specifically, one embodiment is the configuration and interaction of these parts, enabling the pair of opposing alignment features **406** (also referred to herein as protrusions **406**) to pass through the pair of opposing slots **606** and into the pair of opposing channels **408** to allow for moving the cartridge **102** between the stored and dispensing positioned and, ultimately, sliding the vial **602** within the shell **600** for dispensation of the medication.

[0113] The pair of opposing slots **606** are formed through the surface of the shell **600** to allow access to the channels **408** therein. These slots **606** are designed with precise measurements to fit and accommodate corresponding elements, namely, a distinct pair of opposing protrusions **406**. The opposing slots **606** are carved or formed into the shell **600** and aligned in such a way that they maintain a clear, unobstructed path for the protrusions **406** to pass through them with ease. The unique arrangement allows for the insertion of the pair of opposing protrusions **406**, which are another critical characteristic of this one embodiment. These protrusions **406**, with their matching dimensions to the slots **606**, are designed to glide smoothly through the opposing slots **606** and into the aligned opposing channels **408**. The path provided by the slots **606** guides and directs the protrusions **406** into the appropriate channels **408**. Consequently, this ensures a specific interaction between the three parts: the opposing protrusions **406** smoothly pass through the opposing slots **606**, and they are then guided into the opposing channels **408**, maintaining a suitable and secure connection.

[0114] Further, the medication cartridge **102** presents an innovative design with respect to the pair of opposing slots **606**. Each of the pair of opposing slots **606** is formed of a first slot **708**, a second slot **710**, and a third slot **712**. These slots are strategically positioned and oriented for the optimal functioning of the device. The first slot **708** in each pair begins at the bottom end **706** of the shell

600 and extends upward toward the top end **704**. It is arranged in such a way that it mandates a route which ascends from the base to a designated position along the length of the cartridge **102**, otherwise referred to as the first junction **714**.

[0115] Subsequent to the first junction **714**, the structure of the shell **600** features a second slot **710** for each pair. Instead of following the upward trajectory of its predecessor, this second slot **710** takes on a unique characteristic of its own by providing an angled path that descends from the first junction **714**. This downward movement proceeds until it reaches a designated point along the body of the shell **600**, referred to herein as the second junction **716**. Moreover, the slot design of the shell **600** reverts back to an upward direction in the subsequent and final arrangement of each pair—the third slot **712**. Commencing from the second junction **716**, this slot ascends once again (e.g., vertically) towards the top end **704** of the cartridge. The distinct positions and paths of the slots **708**, **710**, and **712** provide a sophisticated layout that enhances the fundamental operations of this medication cartridge **102**. Notably, while the channel **408** terminates at the second junction **716** or channel terminal **718**, the third slot **712** proceeds upwards from the second junction **716**. This differentiation allows for the selective dispensing of the medication therein. This innovative structure enhances the controlled dispensing of medication, making this cartridge **102** particularly advantageous in the administration of drugs. The angles and special slot layout improves the overall efficiency and operational effectiveness of the delivery system, which in turn can broaden its potential applications in the medical and pharmaceutical fields. The precise orientation of the slots from bottom to top and their alternating directions contribute to creating a balanced and smooth transition of the medication within the cartridge **102**. The level of control provided by this inventive structure is a significant evolution in medication delivery system designs.

[0116] For further understanding, FIGS. **8A** and **8B** depict a medication cartridge **102** as being installed or otherwise inserted into the cartridge carrier **200**. As shown, the alignment features **406** (i.e., opposing protrusions) provided a keyed connection and alignment by passing through the channels **408** and slots **606** to allow the cartridge **102** to be contained within the carrier **200**. As shown in FIG. **9**, the alignment feature **406** has travelled through the first slot **708** and channel **408** to rest at the first junction **714**. In this position, the medication cartridge **102** is held within the dispensing device in a stored position. Since the alignment feature **406** is at a fixed location within the cartridge carrier **200**, and because the second slot **710** and channel **408** descend downwards at an angle from the first junction **714**, rotation of the cartridge carrier **200** necessarily forces the medication cartridge **102** upwards within the cartridge carrier **200**. These aspects are further depicted in FIGS. **10A** through **10C**.

[0117] FIG. **10A** depicts the medication cartridge in a stored position. As shown, the alignment feature **416** rests within the first junction **714**. As shown in FIG. **10B**, rotation **1000** (via the gear wheel **404**) of the cartridge carrier **200** forces the alignment feature **416** to traverse through the second slot **710** and channel **408** away from the first junction **714**. In doing so and as shown, the medication cartridge **102** is lifted upwards within the cartridge carrier **200**. Further rotation **1000** of the cartridge carrier **200**, as shown in FIG. **10C**, results in the alignment feature **416** to traverse until it rests within the second junction **716** and the channel **408** terminates at the channel terminal **718**. Notably, when the alignment feature **416** is at the second junction **716**, the medication cartridge **102** is lifted from the dispensing device **100** into the dispensing position. Thus, the journey of the protrusions commences from the first junction **714**, progressing towards the second junction **716**. As a consequence of this carefully guided pathway, the medication cartridge **102** experiences an upward shift. This movement is instrumental in the transition of the medication cartridge **102** from what is denoted as the ‘stored position’ to its final location, the ‘dispensing position’.

[0118] While the cartridge **102** is in the dispensing position, the vial **602** therein is in the expanded state and ready to be compressed to dispense medication. As shown in FIGS. **11A** through **11C**, although the channel **408** in the vial **602** has terminated, the slot in the shell **600** continues upwards

as the third slot **712**. Thus, when the medication cartridge **102** is in the dispensing position, a user can selectively press down on the medication cartridge **102** to cause the vial **602** to slide within the shell **600**. As shown in FIG. **11C**, due to the stepped inner wall **1100** and end of the channel **408**, the vial **602** is held in place by the alignment features **406** and only the outer shell **600** can travel downward when pressed down, causing the shell **600** to slide downward over the vial **602** and allowing the pump assembly to be actuated and spray/dispense the medication. This feature is further illustrated in FIGS. **12A** and **12B**. As shown, pressing downward **1200** on the medication cartridge **102** causes the alignment features **406** to travel upwards through the third slot **712**, thereby forcing the shell **600** downward while holding the vial **602** in place. While the shell **600** is forced downward, it compresses the vial **602** from the expanded to compressed state. The transition from the expanded to compressed state activates the pump assembly to force medication **1202** from the vial **602** and through the dispensing aperture **604**, similar to a squirt of medication. The pump assembly can be configured as a spring-loaded pump mechanism that lifts the shell **600** after being compressed, thereby returning the alignment features **406** to the second junction **716**. Once dispensed, the dispensing device **100** and associated activator mechanism **300** can be activated to turn the gear wheel **404** in an opposite direction as previously rotated. Such a rotation lowers the medication cartridge **102** back to the stored position as the alignment features **406** are forced back to the first junction **714** (shown in FIG. **10A**).

[0119] In one aspect, the device **100** is formed to sense once a predetermined number of squirts (e.g., one, etc., as prescribed) have been initiated so that it automatically activates the activator mechanism **300** to return the medication cartridge **102** back to the stored position, thereby retracting the medication cartridge **102** and inaccessibility after user. For example, magnetic sensors, a trigger, electrical sensors, light sensors, or any other means for determining if medication has been dispensed from the vial **602** can be included within the device **100**. As a non-limiting example, a magnet can be positioned at the bottom of the shell **600** while a magnetic reed switch is similarly positioned at the bottom of the cartridge carrier **200**. Pressing the medication cartridge **102** and its shell **600** downward within the cartridge carrier **200** brings the magnet into close proximity of the magnetic reed switch to notify the associated electronics (integrated circuit, etc.) that a single squirt has been dispensed. After the predetermined number of squirts, the activator mechanism is activated to rotate the cartridge carrier **200** and return the medication cartridge **102** to the stored position to prevent further and unauthorized use. In another aspect, the device **100** can include any necessary components (integrated circuit, etc.) to allow for a timed configuration in which the medication cartridge **102** is held in the dispensing position. For example, the device **100** can be programmed such that after authentication and movement of the medication cartridge **102** into the dispensing position, it is returned to the stored position after a predetermined or preprogrammed amount of time (e.g. 2 seconds, etc.).

[0120] As noted above, a unique design feature allows for the vial **602** to exist in two distinct states; namely an expanded state and a compressed state. The transition between these two stages is critical to the functionality of the device. When it moves from the expanded state towards the compressed state, the incorporated pump assembly **700** is triggered into action. This incorporation of a pump assembly **700** and its associated operation indicates an upbeat sophistication of the one embodiment, going beyond traditional static medical dispensing devices. The purpose of this actuation is specifically to ensure the controlled release of the medicine within the vial **602**. On the actuation of the pump assembly **700**, the medicine **1202** is pushed out of the vial's **602** outlet. This outflow proceeds under the controlling and moderating role of the pump assembly **700**, guaranteeing a controlled and managed release of the medicine.

[0121] In summary, the present disclosure provides an innovative medication dispensing device **100**. The core of this medication dispensing device **100** revolves around a unique cartridge carrier **200** design, integrated with a pair of opposing protrusions **406** and a uniquely designed medication cartridge that is functionally designed to operate based on the positioning of the protrusions **406**

after user authentication. As noted above, in one aspect, the device **100** includes a authentication component **106**. Upon successful authentication, the authentication component **106** triggers the activator mechanism **300**. In this context, the activation entails the initiation of moving the medication cartridge **102**. When activated, it can efficaciously induce the cartridge carrier **200** mechanism to actuate, progressing the inserted medication cartridge **102** from a stored position to a dispensing position. Maximal effectiveness is ensured as the activator mechanism **106** is engendered only after a successful user authentication process, thereby verifying that the medication is dispensed to the appropriate user. This sequence not only ensures secure and controlled dispensing of the medication but also minimizes errors and the potential for unlawful access. In one aspect, the interconnectedness between the authentication component **106** and the activator mechanism **300** forms the crux of this system, thereby augmenting the overall safety and operation of the dispensing device **100**. Moreover, this intricate mechanism, in coordination with the user and the cartridge carrier **200**, ensures that the process of moving the medication cartridge **102** from a stored position to a dispensing position is not only smooth but also efficient and secure. [0122] Thus, in one aspect, a purpose of this inventive operation is to modify the positioning of the medication cartridge **102** from a state of storage to a dispensing state, thereby facilitating the delivery of medication in a controlled, efficient, and timely manner. The operational process involved in use of the dispensing device **100** advances through several stages, starting with the activation of the activator mechanism **300**. The initiation of this mechanism **300** leads to a rotational motion in the cartridge carrier **200**, a critical movement that fosters the maneuvering of the pair of opposing protrusions **406** (i.e., alignment features). This rotation is precisely designed to stimulate this movement, making the entire process efficient and seamless. The pair of opposing protrusions **406** are guided in their motion by a pair of opposing slots **606** in conjunction with a pair of opposing channels **408**. These channels **408** and slots **606** function in harmony, not only providing a safe passage but also controlling the direction of the movement of these protrusions **406** while guiding the medication cartridge **102** between a stored position and a raised dispensing position. The unique features of the slots **606** and channels **408** also allow a user to compress the vial **602** within the medication cartridge **102** to dispense medication **1202**.

[0123] As noted above, the present disclosure provides two example embodiments of the medication cartridge, both of which can be used with the handheld medication dispensing device. While FIGS. 1A through 12B were directed to a first embodiment of the medication cartridge **102**, FIGS. 1B and 13A through 16E depict and are directed to a second embodiment of the medication cartridge **101**. It should be noted that the dispensing device **100** as described herein can be also configured and/or programmed to work with the second medication cartridge **101** according to the necessary features/operations as described herein. Further, in some aspects, each of the features of the first medication cartridge **100** are equally applicable to the second medication cartridge **101** with the exception of the features specifically described below.

[0124] Importantly and as shown in FIGS. 13A and 13B, in addition to all of the components and features as described above, the dispensing device **100** includes the cartridge carrier **200** with a keyed connector (i.e., alignment feature or pair of protrusions **406**) that is designed to align with and operate the medication cartridge **101**. In addition to ensuring the correct placement and alignment of the medication cartridge **101** within the dispensing device **100**, the protrusions **406** are operable for facilitating dispensing/pumping of the medicine within the medication cartridge **101**.

[0125] FIGS. 14A through 14C depict the medication cartridge **101** of the second embodiment. As shown, the medication cartridge **101** includes a shell **1400** that conceals a medicine vial **1402** slidably positioned therein. The proximal or top end **1410** of the shell **1400** includes a pump cap **1404** with a dispensing aperture **1406**. Importantly, the top or proximal end of the vial **1402** includes a pump mechanism **1408** integrated therein that is operable for pumping fluid from the medicine vial **1402** and out of a dispensing aperture **1406**. The pump mechanism **1408** is any suitable mechanism or device that is operable for selectively pumping fluid from the vial **1402** and

out of the dispensing aperture **1406**. As a non-limiting example, the pump mechanism **1408** is similar to as described and illustrated in FIG. 7H. However, instead of pressing down on the medication cartridge **100** as shown in FIGS. **12A** and **12B**, in the second embodiment, the medicine vial **1402** is forced upward within the shell **1400** to compress the pump mechanism **1408** and draw/force medicine or fluid from the vial **1402** and out of the aperture **1406**.

[0126] Compressing the pump mechanism **1408** within the medication cartridge **101** is accomplished with the unique slotting and channeling on the shell **1400** and vial **1402**, respectively. In this aspect, the bottom end **1412** of the shell **1400** has one or more rising slots **1414** (e.g., a pair of opposing slots) that pass through the shell **1400** and rise from the bottom end **1412** toward the top end **1410**. Similarly, the bottom end **1416** of the vial **1402** includes one or more rising channels **1418** (e.g., pair of opposing channels) that are formed within an exterior surface of the vial **1402** and that align with the one or more slots **1414**. Each of the slots **1414** and channels **1418** rise from the bottom ends **1412** and **1416** until they reach a common apex **1422**, at which point they directionally diverge. At the apex **1422**, the rising slots **1414** turn laterally and continue passing through the shell **1400** as horizontal or lateral slots **1424**. Alternatively, at the apex **1422**, the rising channels **1418** alter course and descend at an angle (below the lateral slots **1424**) as downward descending channels **1426**. It should be noted that although the illustrations only depict slots and channels on one side, the invention is not intended to be limited thereto as similar opposing slots and channels can be formed on the other side of the shell **1400** and vial **1402** that engage with the opposing protrusions **406**. Further, each of the downward descending channels **1426** terminate in a channel terminus **1430** which coincides with the end of the pumping motion provided by the pump mechanism **1408**.

[0127] Thus and as was the case above, the rising slots **1414** and lateral slots **1424** are formed through the surface of the shell **1400** to allow access to the channels **1418** and **1426** therein. The slots **1414** and **1424** are formed through the shell **1400** and aligned in such a way that they maintain a clear, unobstructed path for the pair of protrusions **406** to pass through them with ease and into the aligned opposing channels **1418** and **1426**. For example and as shown in FIGS. **15A** and **15B**, the medication cartridge **101** is formed to slide into the dispensing device **100** by aligning the protrusions within the rising slots **1414** and associated rising channels **1418**.

[0128] Once inserted within the dispensing device **100**, the medication cartridge **101** can be selectively activated to pump medicine from the vial **1402**. The dispensing device **100** can be activated using any suitable technique and/or process as described above to cause the motor to activate and, in doing so, rotate the gear wheel and cartridge carrier.

[0129] However and to be contrasted with the design referenced above, in this aspect and as shown in FIGS. **15A** and **15B**, the dispensing device **100** can be modified slightly to include a rotation lock **1500**. The rotation lock **1500** is any mechanism, device, or configuration that allows for rotation of the cartridge carrier **200** while preventing rotation of an inserted medication cartridge. As a non-limiting example, the rotation lock **1500** can be a protrusion or other form or marking that is attached within the dispensing device **100** and protrudes upward into the cartridge carrier **200** to lock against a portion of the medication cartridge **101** (e.g., against the shell and/or vial). Alternatively and as yet another example, the rotation lock **1500** can be a depression formed within the dispensing device and below the cartridge carrier **200** that engages with a protrusion that projects from the bottom of the medication cartridge **101**. In one aspect and as shown in FIG. **13B**, a hole **1300** or other aperture can be formed in the bottom of the cartridge carrier **200** (and associated gear wheel) to allow the rotation lock **1500** to pass therethrough and lock against a portion of the medication cartridge **101**. In this aspect and as shown in FIG. **14A**, a keyed marking **1440**, such as an indentation or other shaping, can be formed in the bottom of the medication cartridge **101** (e.g., on the bottom exterior or bottom side exterior of the shell **1400**) to allow the rotation lock **1500** (of FIGS. **15A** and **15B**) to lock against the keyed marking **1440** and prevent rotation therebetween.

[0130] Further, a slidable coupling is included to prevent rotation between the shell **1400** and vial **1402**. The slidable coupling is any suitable mechanism, device, or configuration in which the vial **1402** is prevented from rotating with respect to the shell **1400**, yet allowed to slide within the shell **1400**. As a non-limiting example, protruding inward from the interior walls of the shell **1400** are a pair of elongated ridges **1442** while the exterior wall of the vial **1402** includes an elongated channel **1444**. When the vial **1402** is within the shell **1400**, the ridges **1442** align and slide within the elongated channel **1444**, yet prevent rotation between the vial **1402** and shell **1400**. Thus, in this aspect, locking of the lock **1500** against the keyed marking **1440** prevents rotation of the medication cartridge **101** itself when the cartridge carrier **200** is actuated and caused to rotate, while still allowing the vial **1402** to slide vertically within the shell **1400**.

[0131] As shown between FIGS. **16A** and **16E**, rotation of the cartridge carrier **200** in a first direction causes the vial **1402** to rise within the shell **1400** from a stored position (FIG. **16A**) to a full pump position (FIG. **16C**) and activate the pump mechanism **1408**, thereby dispensing medicine from the medication cartridge. This process is enabled by the unique slots, channels, and keyed marking.

[0132] As shown in FIG. **16A**, when the medication cartridge **101** is positioned down and into the dispensing device and associated cartridge carrier **200**, the protrusions **406** pass through the rising slots **1414** and rising channels **1418** until the protrusions **406** come to rest at the apex **1422**. In use, activation of the dispensing device causes motor to activate and rotate the cartridge carrier **200** in the first direction. As shown in FIG. **16B**, as the cartridge carrier **200** rotates, the protrusions **406** turn laterally and pass through the lateral slots **1424**. Notably, because the downward descending channels **1426** are angled downward and away from the apex **1422**, continued rotation of the protrusions **406** within the descending channels **1426** causes the vial **1402** to rise upwards **1600**. Forcing the vial **1402** upwards compresses the pump mechanism **1408** to activate the pump mechanism **1408** and force fluid or medicine from the vial **1402**. As shown in FIG. **16C**, the cartridge carrier **200** continues to rotate and, in doing so, the protrusions **406** continue to slide within the descending channels **1426** and force the vial **1402** up to compress and activate the pump mechanism **1408**. This process continues until the protrusions **406** reach the channel terminus **1430**, which coincides with the end of the pumping motion provided by the pump mechanism **1408**.

[0133] In one aspect, upon completion of the pump cycle, the dispensing device reverses motion of the motor which causes the cartridge carrier **200** to rotate in the reverse direction (i.e., a second direction). As shown in FIGS. **16D** and **16E**, as the cartridge carrier **200** reverses rotation, the protrusions **406** slide within the descending channels **1426** to force the vial **1402** downwards **1602** until the protrusions reach the apex **1422**. Upon reaching the apex **1422**, the vial **1402** has returned to the stored position and is ready to be activated again in another pump/dispense cycle.

[0134] Finally, while this invention has been described in terms of several embodiments, one of ordinary skill in the art will readily recognize that the invention may have other applications in other environments. It should be noted that many embodiments and implementations are possible. Further, the following claims are in no way intended to limit the scope of the present invention to the specific embodiments described above. In addition, any recitation of “means for” is intended to evoke a means-plus-function reading of an element and a claim, whereas, any elements that do not specifically use the recitation “means for”, are not intended to be read as means-plus-function elements, even if the claim otherwise includes the word “means”. Further, while particular method steps have been recited in a particular order, the method steps may occur in any desired order and fall within the scope of the present invention.

Claims

- 1.** A medication cartridge, comprising: a shell having a dispensing aperture; and a vial positioned in the shell, the vial being movable between a stored position and a full pump position to dispense medicine from the vial and dispensing aperture.
- 2.** The medication cartridge as set forth in claim 1, wherein the shell includes one or more rising slots passing through the shell and rising from a bottom end of the shell toward an apex.
- 3.** The medication cartridge as set forth in claim 2, wherein the vial includes one or more rising channels that align with the one or more rising slots and rise from a bottom end of the vial toward the apex.
- 4.** The medication cartridge as set forth in claim 3, wherein the shell includes one or more lateral slots that traverse laterally away from the apex.
- 5.** The medication cartridge as set forth in claim 4, wherein the vial includes downward descending channels that descend downward at an angle from the apex.
- 6.** The medication cartridge as set forth in claim 5, wherein the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.
- 7.** The medication cartridge as set forth in claim 6, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.
- 8.** The medication cartridge as set forth in claim 7, wherein the vial includes a pump mechanism with an outlet aligned with the dispensing aperture, the vial being slidable within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the vial rises within the shell to actuate the pump mechanism to pump medicine from the vial and out of the dispensing aperture.
- 9.** The medication cartridge as set forth in claim 8, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.
- 10.** The medication cartridge as set forth in claim 1, wherein the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.
- 11.** The medication cartridge as set forth in claim 10, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.
- 12.** The medication cartridge as set forth in claim 1, wherein the vial includes a pump mechanism with an outlet aligned with the dispensing aperture, the vial being i-s slidable within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.
- 13.** The medication cartridge as set forth in claim 1, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.
- 14.** The medication cartridge as set forth in claim 1, wherein the vial is slidably positioned within the shell between the stored and full pump position, the vial having a pump mechanism with an outlet aligned with the dispensing aperture and being operable for rising within the shell to activate the pump mechanism and dispense medicine from the vial and dispensing aperture when moved to the full pump position.

15. The medication cartridge as set forth in claim 14, wherein the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.

16. The medication cartridge as set forth in claim 15, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.

17. The medication cartridge as set forth in claim 16, wherein the vial is slidable within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.

18. The medication cartridge as set forth in claim 17, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.
